Outliers United : 2025 Tariff

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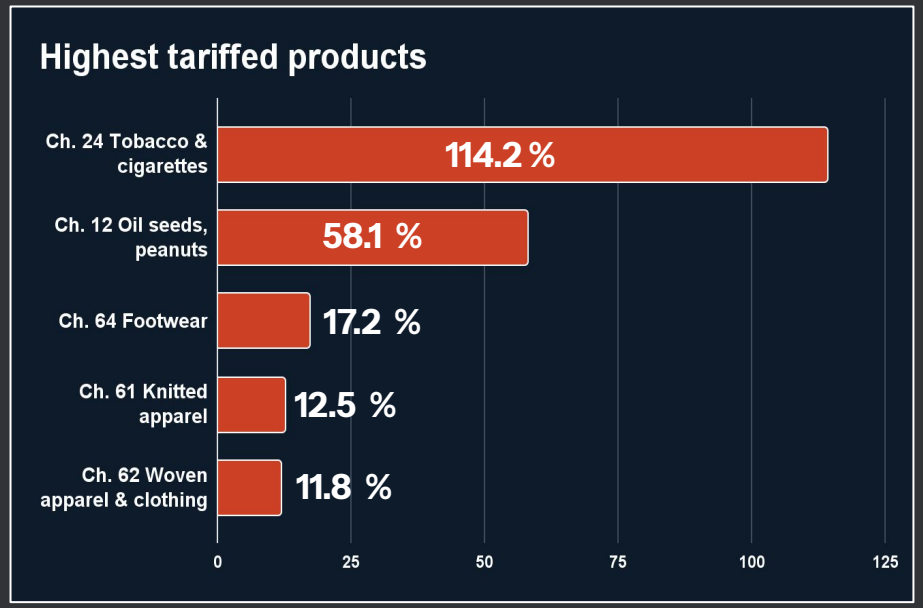
Project Github Repo:
2025 HTS

2025 Harmonized Tariff Schedule

13,058 Product Codes	98 Chapters/Groups of products	35% Duty Free Items	Built over 56 years

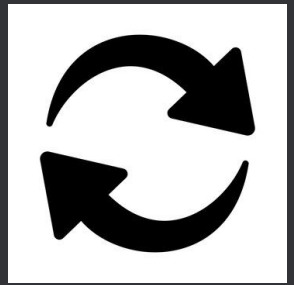
- **HTS Code**
  - Unique 8 digit ID for every product
- **Product description**
  - Plain english description
- **MFN Rate**
  - base tax rate for all countries
- **19 FTA**
  - Preferential tax for friendly countries
- **Col. 2 Rate**
  - Penalty for non favored countries



Intention

Goals:

- Audit the structural consistency of the 2025 Harmonized Tariff Schedule (HTS)
- Model production supply chains and compare the costs of importing raw goods vs. finished products
- Identify cases where raw materials are taxed greater than manufactured goods (inversions)
- Determine if the 2025 policy inadvertently penalizes domestic manufacturing by creating negative effective protection
- Analyze how tariff amount across industries change over time and what drives those changes



Stakeholders and Their Uses

Domestic Manufacturers:

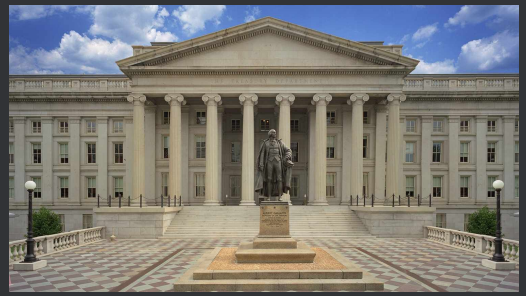
- See how tariffs on raw material imports impact production costs
- Determine what portion of tariff costs can be passed on to the consumer

Government Entities:

- Track effectiveness of tariffs in improving domestic production
- Understand how tariffs affect costs for consumers

Retail Stores:

- Compare cost of sourcing inventory domestically vs. internationally



Supply Chain Tariff Inversions

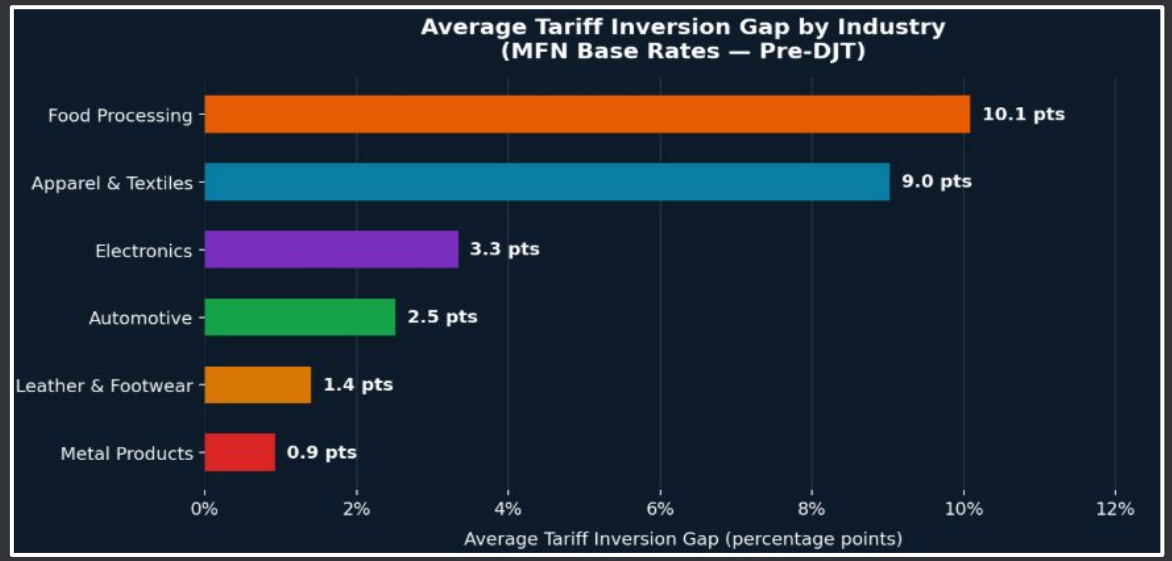
- Clean 2025 HTS with 13058 product codes
- Directly linked specific raw materials to specific finished goods then compared their rates head to head.
  - Used keyword matching on HTS descriptions
  - Raw material → Finished product
  - Across 6 industries 682 inverted pairs
  - Verified the matchings with BEA Input Output by Industry

To make it at home ?
Or to import ?

Supply Chain Tariff Inversions

682 Inverted supply chain pairs

6 Industries



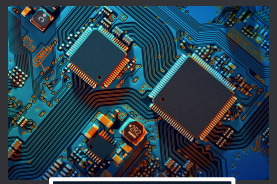
26.4%

1.4%

6.5%

1 %

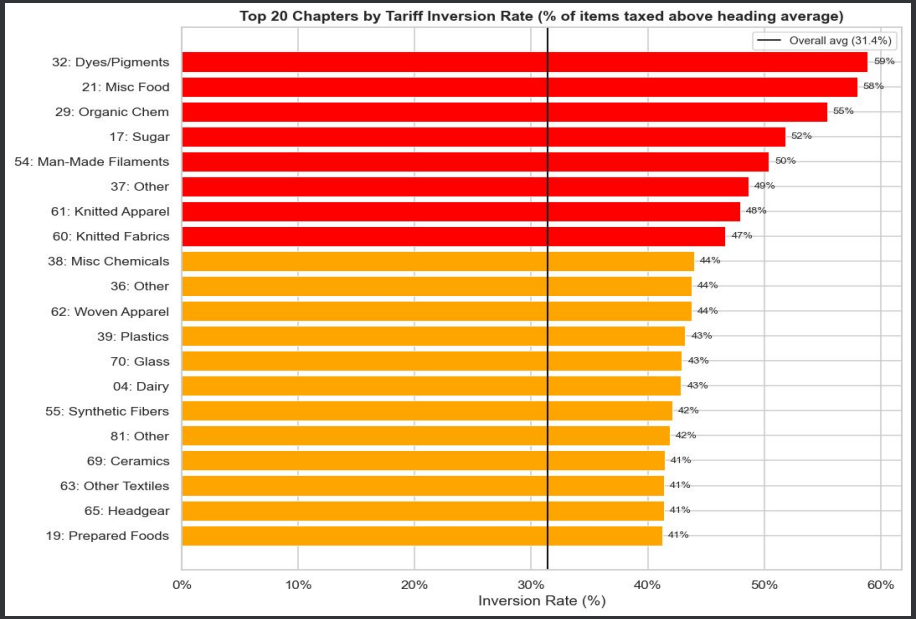
3 %



0 - 2.6 %

Tariff Inversion & FTA Relief Analysis

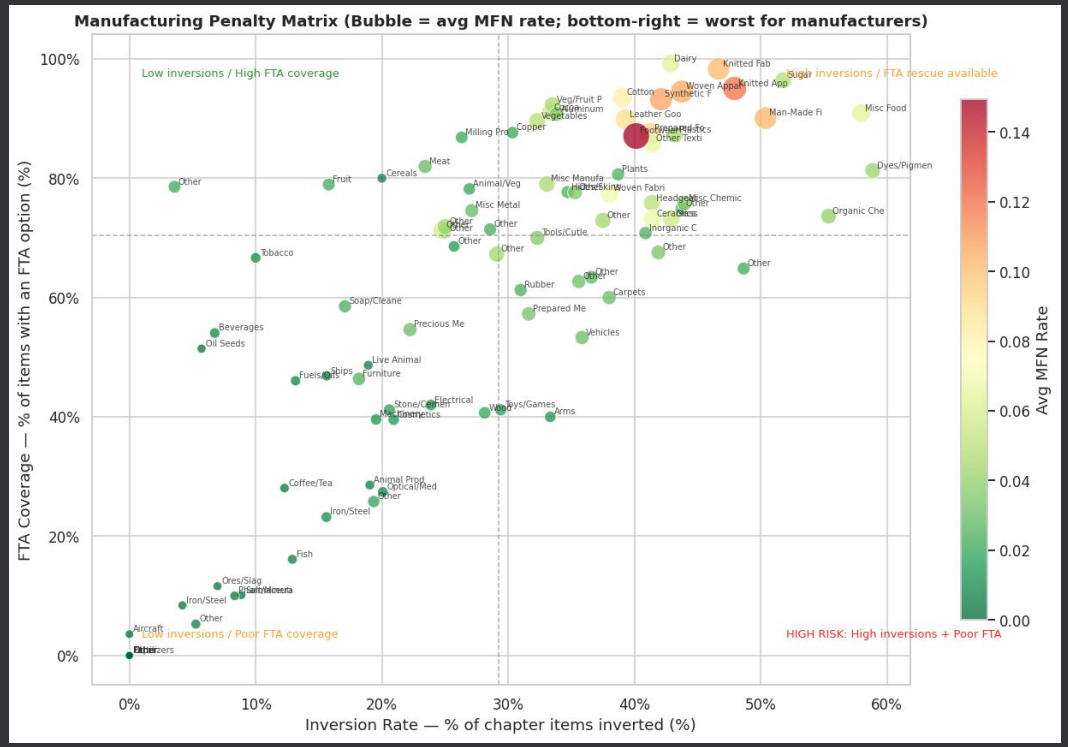
Research Question: To what extent do Free Trade Agreement preference programs offset tariff inversions in the 2025 HTS, and which industries remain structurally penalized regardless of available relief?



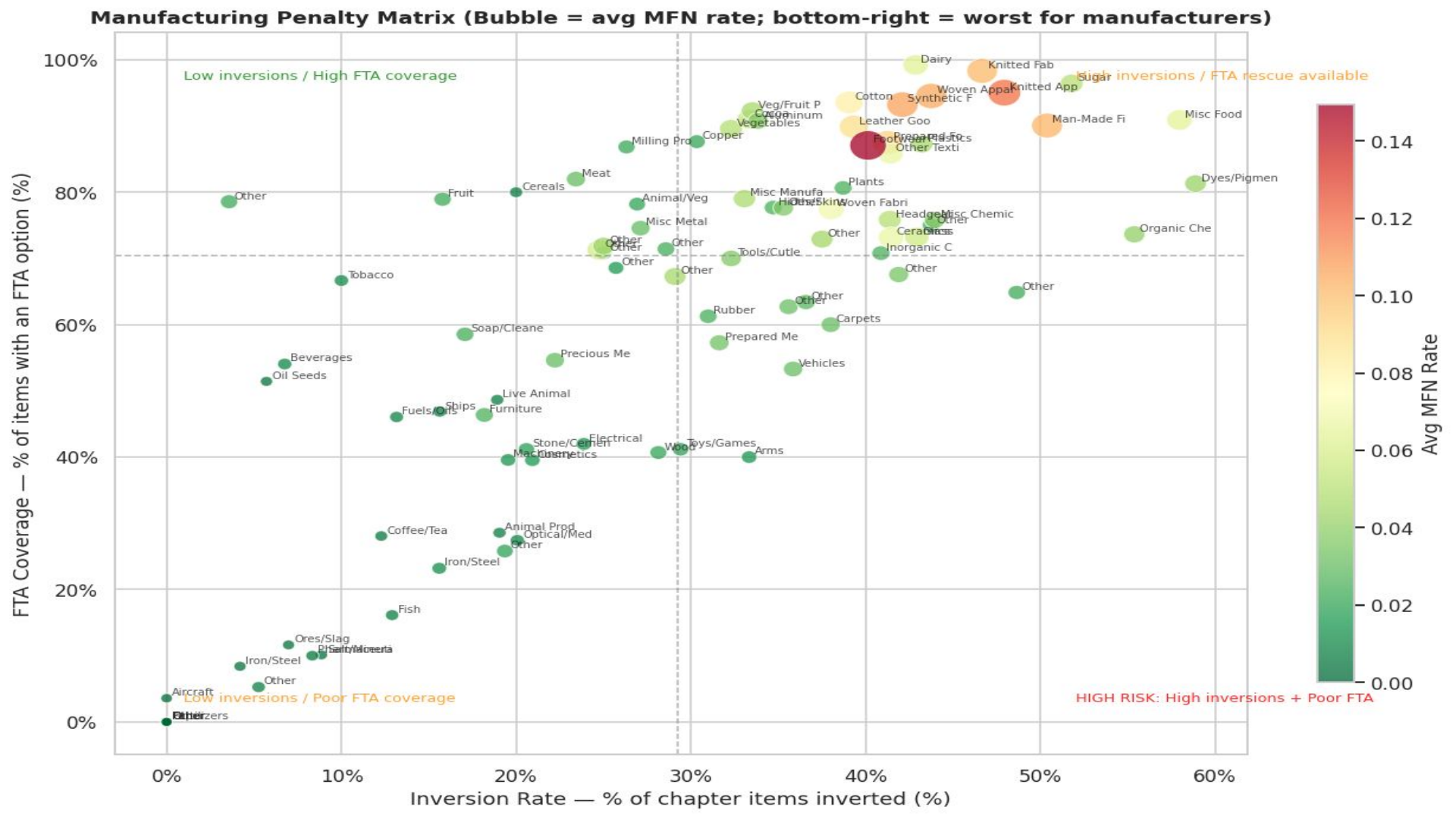
Analytical Approach:

- Flagged tariff inversions by comparing each item's MFN rate against its heading average
- Mapped all 19 FTA/preference program rate columns to compute the best available relief rate per item, then calculated the gap between MFN rate and effective rate after FTA
- Combined both dimensions into a manufacturing penalty matrix plotting inversion rate vs. FTA coverage at the chapter level to identify structurally penalized industries

Findings



- 36% of HTS items exhibit a tariff inversion, with the highest rates (50%+) in Dyes/Pigments, Organic Chemicals, Sugar, and Man-Made Filaments
- FTA programs nearly fully eliminate tariff burdens for the highest-tariff chapters, covering ~62% of inverted items
- Avoiding the manufacturing penalty relies on supply chain flexibility to source from FTA partner countries which is not a policy gap


Manufacturing Penalty Matrix (Bubble = avg MFN rate; bottom-right = worst for manufacturers)
Low inversions / High FTA coverage
High inversions / FTA rescue available
Low inversions / Poor FTA coverage
HIGH RISK: High inversions + Poor FTA
FTA Coverage — % of items with an FTA option (%)
Inversion Rate — % of chapter items inverted (%)
Avg MFN Rate
100%
80%
60%
40%
20%
0%
0%
10%
20%
30%
40%
50%
60%
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
Dairy
Knitted Fab
Sugar
Knitted App
Woven Appar
Synthetic F
Cotton
Veg/Fruit P
Aluminum
Vegetables
Leather Goo
Man-Made Fi
Misc Food
Copper
Other Texti
Meat
Plants
Dyes/Pigmen
Other
Fruit
Cereals
Animal/Veg
Misc Manufa
Woven Fabri
Misc Chemic
Misc Metal
Organic Che
Other
Tools/Cutle
Inorganic C
Tobacco
Rubber
Carpets
Soap/Cleane
Prepared Me
Beverages
Precious Me
Vehicles
Oil Seeds
Live Animal
Ships
Furniture
Electrical
Toys/Games
Arms
Coffee/Tea
Animal Prod
Optical/Med
Iron/Steel
Fish
Ores/Slag
Iron/Steel
Other
Aircraft

Tariff Wall Similarity

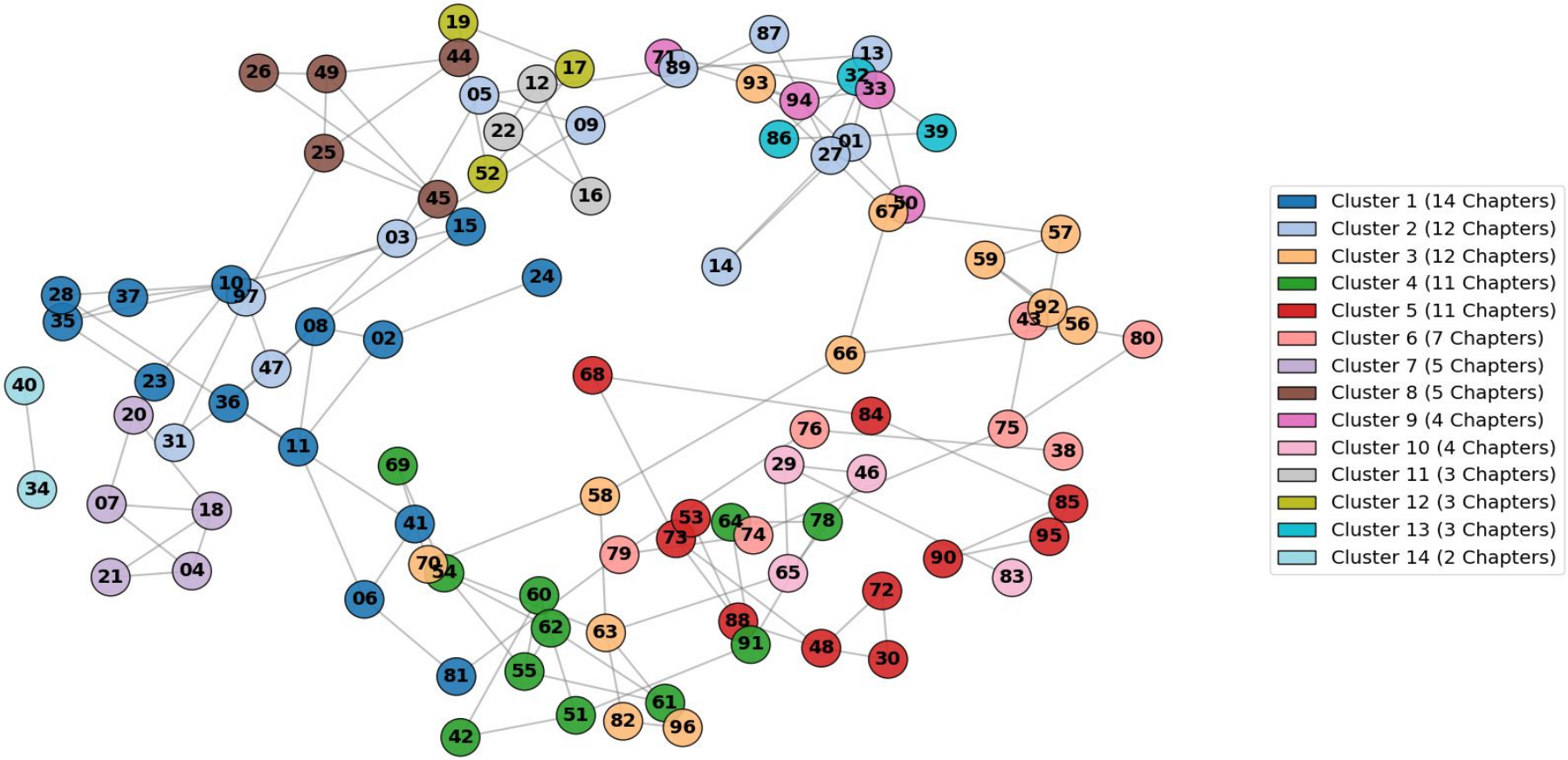
Methodology

1. Computed 3 chapter-level features: avg MFN rate, avg Col 2 rate, % duty-free items
2. Standardized features with z-score scaling for fair comparison
3. Built a cosine similarity matrix across all 97 chapters
4. Constructed a graph using similarity matrix
5. Applied greedy modularity community detection to find clusters

Cluster Findings

1. Ch. 50–67: All apparel/fabric chapters share extremely high MFN & Col 2 rates
2. Ch. 1–24: Food & farming chapters share moderate MFN but high penalty rates
3. Ch. 25–40: Chemicals, plastics, minerals are mostly low-tariff or duty-free
4. Ch. 84–92: Electronics/machinery — low MFN but selective Col 2 protection

'Tariff Wall' Similarity Network
(Chapters clustered by similar tax/tariff structures.)



Preference Exclusion Hypothesis

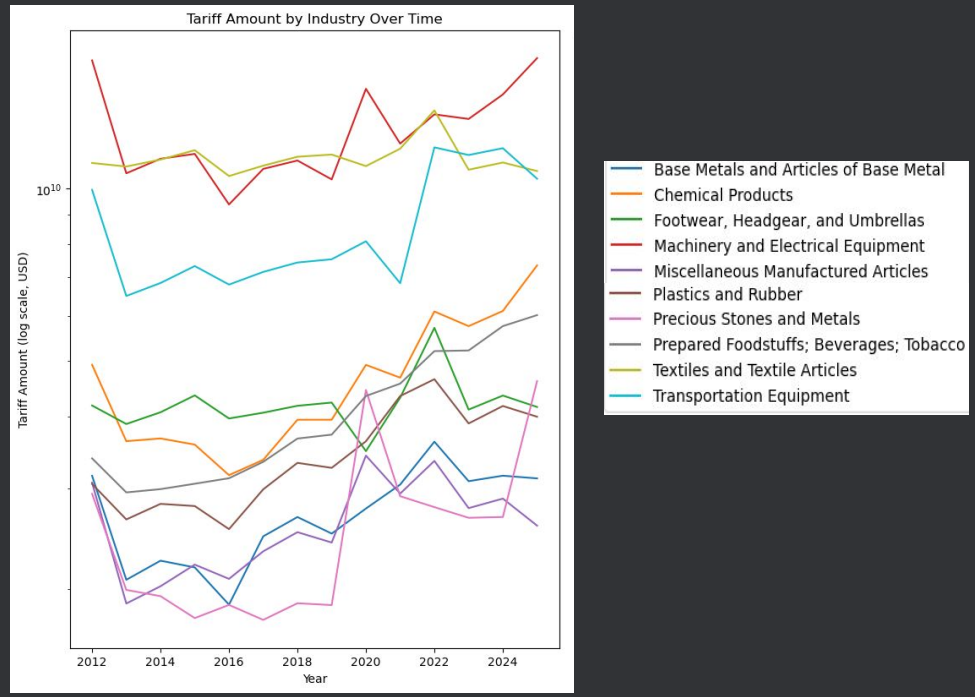
Hypothesis Statement: "The U.S. is most likely to exclude products from GSP/AGOA free-trade programs if those products are already protected by a high domestic tariff wall."

tariff_wall	Total_Products	Avg_MFN_Rate	Percent_Granted_GSP	Percent_Granted_AGOA
Duty-Free (0%)	5073	0.000000	12.240000	12.730000
Low Tariff (0-5%)	2793	0.030000	91.050000	91.510000
High Wall (>5%)	3383	0.100000	55.960000	60.720000

Conclusion: The data supports the Preference Exclusion Hypothesis. U.S. trade preferences follow a 'conditional generosity' model where free access is granted freely only where domestic manufacturers are not threatened.

Tariff Share Time Series Analysis

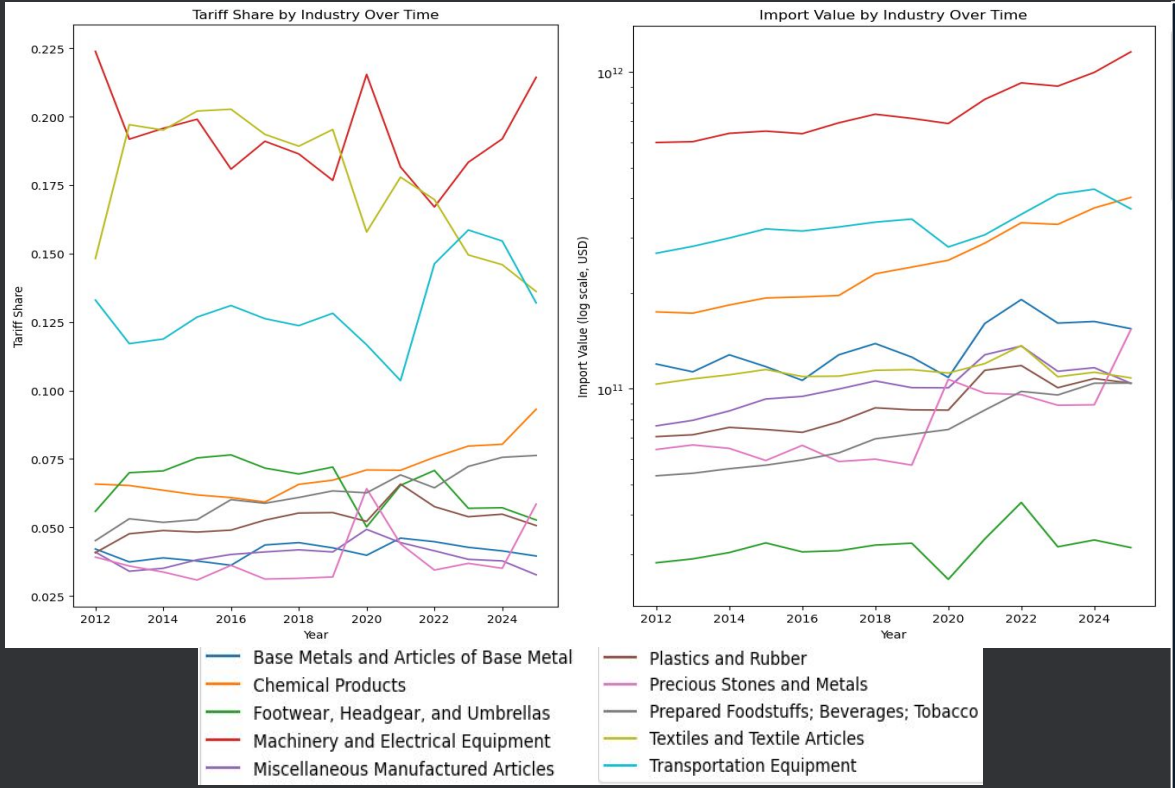
Research Question: Which industries account for the highest tariff share over the past several years and how has changes in import value affected tariff share?



Analytical Approach:

1. Collect annual tariff schedule and annual import data by HTS chapter
2. Format HTS codes, align chapters across the datasets and remove placeholder values
3. Map HTS Chapters to industry groups as grouped on Tariff Schedule website
4. Compute average tariff rate by chapters and merge with chapter level import value by year
5. Calculate tariff value per chapter by multiplying tariff rate by import value
6. Calculate and visualize the top 10 tariff values by industry (log transformed)
7. Calculate and visualize industry share of tariff imports by calculating the tariff value per industry divided by total tariff value

Tariff Share Time Series Analysis



Observations:

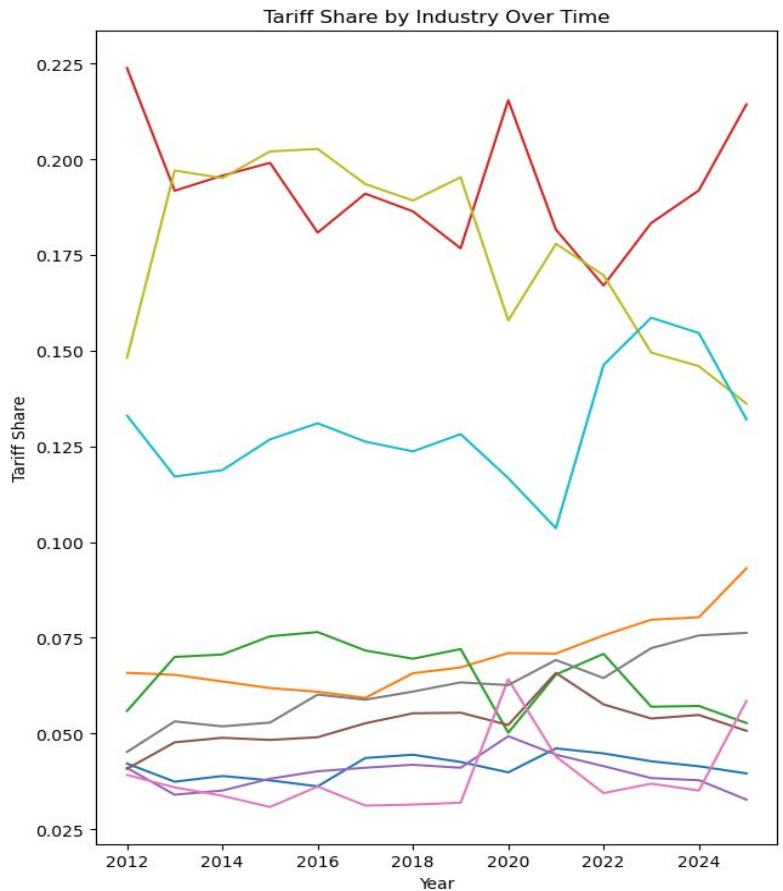
- Tariff amounts are concentrated with the top three industries accounting for 50% of the total tariff share
- Overall tariff amounts and tariff value have increased in recent years
- Distribution of tariff share remains stable

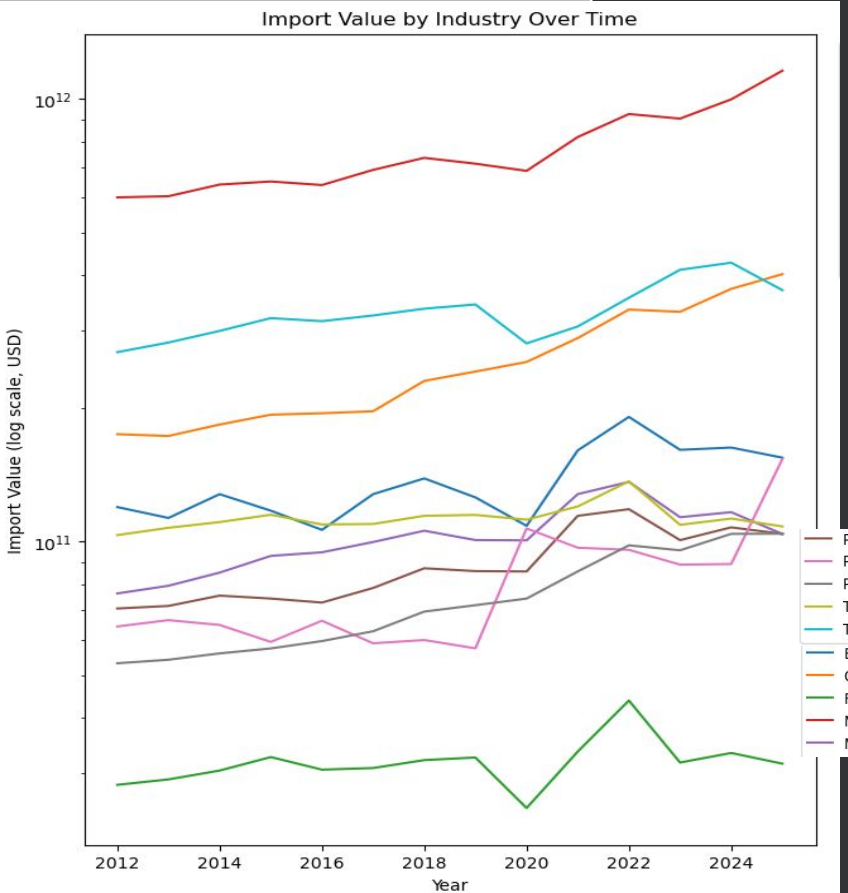
Conclusion:

- Recent increase in tariff amount are driven by the rising import values.
- Suggests changes in trade volume and demand have had a greater impact on tariff amount than changes in tariff rates themselves.

Important Trade Events:

- *2018–2019:* US–China trade war
- *2020:* Global trade contraction during COVID-19
- *2021–2023:* Post-pandemic recovery in import





Challenges, Limitations and Future Work

- Keyword matching has blind spots. Products with no shared keywords are invincible and real supply chain could be missed
- Changing tariff policies make it hard it to create timely analysis
- Rate outliers in MFN column (large placeholder) threw off rate calculations at and some FTA rates contained 0.0 values (distinguishing duty-free vs. no coverage)
- Amounts not being mentioned require another dataset to be used in conjunction
- Import value data doesn't identify individual imports, so duty free imports (below $800) cannot be identified

- Analyze import volume data to weight each industry inversions for each product.
- Expand finished supply chain pairs to account for all finished goods that are coming into the US.
- Track inversions over time starting from 1990 accounting for Bilateral trade deals, FTAs, WTO policies.
- Compare U.S inversions to trade partners with highest tariffs and lowest tariffs.
- Model consumer impact and calculate industry damage overtime.

Team Contributions

Ari Munkhtur
- Tariff inversions
- Raw - Finished

Ashay Patla
- Tariff Share Analysis
- Time Series Visualization

Jayesh Bane
- Tariff Wall Similarity
- Preference Exclusion Hypothesis

Ryan Quinlan
- Tariff Inversion Analysis
- Tariff Refund Analysis

Luke Ricciardi
- Raw Data Analysis
- Industry Application Research

Thank You